Food Chemistry 374 (2022) 131630

Contents lists available at ScienceDirect

Food Chemistry

journal homepage: www.elsevier.com/locate/foodchem

Carvone and its eutectic mixtures as novel, biodegradable, and tunable solvents to extract hydrophobic compounds in substitution for volatile toxic solvents

Ke Li [a], Jin Hyun Nam [b], Seulgi Kang [a], Yuli Liu [a], Jeongmi Lee [a,*]

[a] School of Pharmacy, Sungkyunkwan University, Suwon 16419, Republic of Korea
[b] Department of Big Data Science, Korea University-Sejong Campus, Sejong 30019, Republic of Korea

ARTICLE INFO

Keywords:
Carvone
Bio-based solvent
Eutectic mixture
Extraction
COSMO-RS
Machine learning

ABSTRACT

Finding safe solvents with low viscosities has been in great demand in extraction processes. Herein, R-(-)-carvone, a natural monoterpenoid rich in spearmint, was mixed with naturally occurring fatty acids and terpenes. Most eutectic mixtures presented a wide liquid window in the solid-liquid equilibrium phase diagrams. Carvone mixtures at the ideal eutectic points were characterized for physicochemical properties. Despite varying properties, all the tested solvents were immiscible with water and displayed low viscosity with reasonable biodegradability. Sigma potentials of the mixtures were applied to machine learning algorithms, suggesting carvone mixtures as substitutes for polar protic and dipolar aprotic solvents. Carvone mixtures could be successfully applied to liquid-liquid extraction of a red algae called laver, which is rich in natural hydrophobic and hydrophilic pigments of high value. This study proposes carvone as a new bio-based source of hydrophobic solvents and the eutectic mixtures as biodegradable and tunable solvents to extract hydrophobic compounds.

1. Introduction

Solvents comprise the core of chemical processes in food, pharmaceutical, cosmetic, and agrochemical process technologies. Solvents that fulfill the principles of green chemistry are in great demand, in accordance with massive awareness of and concerns regarding the environment and health impact of the solvents used. Continuous efforts have been made to search for alternative solvents with low pollution, low cost, and low toxicity, to replace conventional petroleum-based solvents in chemical processes. Bio-based solvents such as glycerol, terpenes, and their derivatives, are one type of innovative solvent that may fulfil several criteria of green solvents (Gu & Jerome, 2013). Supercritical fluids, ionic liquids (ILs), and deep eutectic solvents (DESs) have also been actively investigated as potential green solvents (Clarke, Tu, Levers, Bröhl, & Hallett, 2018). DESs can be classified as a sub-type of a eutectic system, in which the eutectic formation significantly decreases the melting point of the mixture in comparison to those of the pure compounds and their ideal mixtures (Martins, Pinho, & Coutinho, 2019). DESs are usually considered superior alternatives to ILs, based on their biodegradability, easy preparation, and low cost (Hansen, Spittle, Chen, Poe, Zhang, Klein, et al., 2021). Eutectic mixtures, whether they are deep or simple eutectic solvents, usually consist of natural hydrophilic compounds, including sugars, sugar alcohols, amino acids, organic acids, and choline derivatives (Hansen, et al., 2021). The instability of these solvents in water or their water miscibility has limited their applications in liquid-liquid extractions (LLE) involving water (Lee, Jung, & Park, 2019).

Glycerol is a water-miscible, bio-based solvent with various desirable properties, including low price, renewability, non-flammability, and biocompatibility. The high dissolving capability of a variety of organic and inorganic compounds makes glycerol an efficient extraction solvent. However, glycerol has two main drawbacks: high viscosity (1,410 mPa·s) (Ozturk, Esteban, & Gonzalez-Miquel, 2018) and low solubility for highly hydrophobic compounds (Garcia, Garcia-Marin, & Pires, 2014). One way to overcome these shortcomings is to mix glycerol with various compounds capable of hydrogen-bonding interactions to form eutectic mixtures (designated as DESs) (Garcia, Garcia-Marin, & Pires, 2014). The solvent properties of glycerol mixtures or glycerol-based DESs, including polarity and viscosity, can be altered, in contrast to pure glycerol (Abbott, Harris, Ryder, D'Agostino, Gladden, & Mantle, 2011). For instance, in a DES composed of glycerol and choline chloride at a 2:1 molar ratio, the viscosity was dramatically reduced by a factor of

* Corresponding author.
E-mail address: jlee0610@skku.edu (J. Lee).

https://doi.org/10.1016/j.foodchem.2021.131630
Received 23 September 2021; Received in revised form 12 October 2021; Accepted 15 November 2021
Available online 19 November 2021
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three (Abbott, Harris, Ryder, D'Agostino, Gladden, & Mantle, 2011). However, glycerol mixtures are still polar and readily miscible with water; their substitution for water-immiscible organic solvents is not plausible.

Several hydrophobic natural terpenes and their derivatives (e.g., limonene, α-pinene, β-pinene, and *p*-cymene) are regarded as bio-based solvents to replace non-polar petroleum-based solvents (Boutekedjiret, Vian, & Chemat, 2014). Among them, limonene, which is a natural oil abundant in many renewable feedstocks such as citrus waste, has been studied the most. Recently, terpenes have been adopted in the preparation of hydrophobic eutectic solvents, including menthol (Ribeiro, Florindo, Iff, Coelho, & Marrucho, 2015), thymol (Martins, Crespo, Pontes, Silva, Bulow, Maximo, et al., 2018), and limonene (Pereira, Silva, Rodrigues, Reis, Paiva, Duarte, et al., 2019). The resulting terpene-based DESs possess significantly reduced viscosity when compared to hydrophilic DESs, as Coulombic charge interactions are absent (Florindo, Branco, & Marrucho, 2019). These solvents have attracted increased attention and form one of the major classes of hydrophobic DESs (Lee, Jung, & Park, 2019). The selection and design of diverse hydrophobic solvents from natural sources are still a focus of the industry, in order to widen the application scope of hydrophobic green solvents for green chemistry.

In our continued efforts to broaden the pool of hydrophobic DESs from natural resources, a natural terpenoid, (R)-(-)-carvone (hereafter referred to carvone) was identified as a potential candidate. Carvone is a monoterpenoid that is a dominant constituent (60%–70 %, *v/v*) in the essential oil of spearmint (*Mentha spicata* L.) (de Carvalho & da Fonseca, 2006) with a low melting point (247.7 K) (Gallis, Bougrioua, Oonk, vanEkeren, & vanMiltenburg, 1996). It has various biological activities, such as antimicrobial, chemopreventive, and anti-tumor effects, with low allergenicity and toxicity (de Carvalho & da Fonseca, 2006). It has been commercially used as a fragrance for hygienic and cosmetic products and has been applied as a monomer feedstock for the synthesis of functional bio-renewable polyesters (de Carvalho & da Fonseca, 2006). Carvone can be obtained directly from spearmint oil and can be synthesized from (R)-(+)-limonene (Lange, 2015). Despite its various applications, carvone has never been studied as a solvent.

In this study, carvone will be investigated as a bio-based solvent that might form a series of eutectic mixtures, with diverse compounds with different properties. In this context, carvone was mixed with three different classes of compounds (fatty acids, terpenes, and a benzoate ester) at various molar ratios, and the liquid window of the eutectic mixtures was identified from the solid–liquid equilibrium (SLE) phase diagrams. The physicochemical properties of the eutectic mixtures were characterized and compared with those of carvone. Their equilibrium properties, which were calculated using a thermodynamic model, COnductor-like Screening MOdel for Real Solvents (COSMO-RS) (Klamt, 2005), were used to place these mixtures among various classes of conventional organic solvents. After biodegradability assessment, the carvone mixtures were applied to the liquid-liquid partition of a model system, composed of a hydrophilic and a hydrophobic dye. Finally, their application to LLE in real samples was performed on a well-known red alga, laver, which is rich in a variety of functional components, including hydrophobic pigments, carotenoids and chlorophylls, and hydrophilic phycobiliproteins, phycocyanin (PC) and phycoerythrin (PE) (Pangestuti & Kim, 2011).

2. Materials and methods

2.1. Materials and preparation of carvone-based mixtures

R-(-)-Carvone and its counterpart compounds in the formation of eutectic mixtures, including heptanoic acid, octanoic acid, nonanoic acid, decanoic acid, (±)-camphor, (±)-menthol, (-)-borneol, thymol, and phenyl salicylate (salol) were purchased from Sigma-Aldrich (St. Louis, MI, USA). Hereafter, the stereochemistry of the compounds is not indicated. They were used as received, without further purification. Detailed information on these compounds is provided in Table 1. Other chemicals including hexane, chloroform, toluene, peptone from milk solids, hydrochloric acid (37%), Nile red, rhodamine 6G, and Brilliant blue FCF were also obtained from Sigma-Aldrich unless otherwise stated. Laver (*Porphyra sp.*) was purchased from a local market in South Korea. HPLC-grade water and methanol were obtained from J. T. Baker (Suwon, South Korea).

A series of eutectic mixtures were prepared by mixing carvone with other compounds at various molar ratios (Table S1), including those listed in Table 2. The mixtures were heated at 60 °C until a clear liquid was obtained, then cooled to room temperature.

2.2. Thermodynamic modelling using COSMO-RS

COSMO-RS calculations were conducted according to previous studies (Klamt, 2005). Predictive solutions were achieved through a combination of COSMOtherm with the Turbomole quantum chemistry suite (BIOVIA COSMOtherm, Release 2021; Dassault Systèmes, Leverkusen, Germany) at the density functional theory (DFT) functional B88-PW86 with a def-triple-z valence polarized basis (TZVP) level. The charge density σ was used for the thermodynamic calculation of the interaction energy, including misfit, hydrogen bonding, and van der Waals forces. Sigma-profile for compound *i*, $p_i(\sigma)$, which is the probability of finding a surface segment with a charge density σ, is defined as

$$p_i(\sigma) = \frac{A_i(\sigma)}{A_i} \tag{1}$$

where $A_i(\sigma)$ is the surface area with charge density σ. Sigma-profile of mixtures, $p_S(\sigma)$ was calculated as the sum of σ-profiles of components using the following equation:

$$p_S(\sigma) = \sum_{i \in S}[x_i p_i(\sigma)] \tag{2}$$

where x_i is the mole fraction of component *i*. Sigma-potential, $\mu_S(\sigma)$ is the chemical potential of a surface segment with σ in solution S, and is a measure of the affinity of system *S* to the surface of polarity σ. This was calculated using Eq. (3):

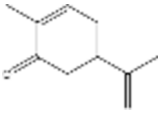
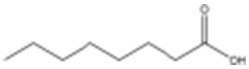
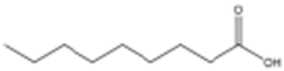
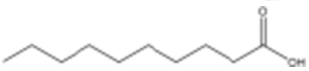
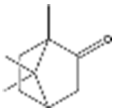
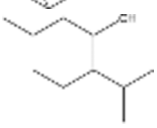
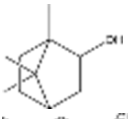
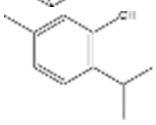
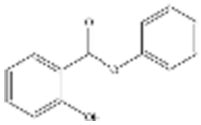
$$\mu_S(\sigma) = -\frac{RT}{a_{eff}} \ln\left[\int p_s(\sigma') e^{\left(\frac{a_{eff}}{RT}[\mu_S(\sigma') - E(\sigma,\sigma')].\right)} d\sigma\right] \tag{3}$$

where $E(\sigma, \sigma')$ is the total interaction energy, that is, the sum of the misfit energy, hydrogen bonding, and van der Waals interactions of two adjacent screening charges, σ and σ', while a_{eff} is the effective contact area. In this way, the σ-profiles and σ-potentials of carvone and nine carvone-based mixtures were calculated. Moreover, 150 classical solvents that have been previously studied for solvent classification (Durand, Molinier, Kunz, & Aubry, 2011) were also calculated. The temperature was set to 25 °C for all calculations.

2.3. Machine learning for clustering and classification

The original dataset for each solvent contained σ-potentials at charge densities from −0.030 e/Å^2 to 0.030 e/Å^2 with an interval of 0.001 (61 variables in total) and was scaled by root mean squares. Data scaling and analysis were performed using R (R Core Team 2020, R Foundation for Statistical Computing, Vienna, Austria, https://www.R-project.org/). The APClust package was used for the clustering of the 150 classical solvents, using the affinity propagation (AP) algorithm. The AP algorithm took the similarity between solvents as input, with similarity being a negative Euclidean distance (Frey & Dueck, 2007). Based on the clustering results from the AP algorithm, carvone and nine additional mixtures were subjected to classification. Classification of carvone and its mixtures was conducted using support vector machines (SVMs) with type of C classification provided by the kernlab package. Clustering

Table 1
Compounds used to prepare eutectic mixtures in this study.

Compound	CAS No.	Chemical structure	Melting temperature (K)
R-(−)-Carvone [a]	6485-40-1		247.7 [b]
Heptanoic acid	111-14-8		262.65[c]
Octanoic acid	124-07-2		288.15–290.15[c]
Nonanoic acid	112-05-0		282.15[c]
Decanoic acid	334-48-5		300.15–305.15[c]
(±)-Camphor	76-22-2		448.15–450.15[c]
(±)-Menthol	88-78-1		307.15–309.15[c]
(−)-Borneol [d]	464-45-9		480.6 [e]
Thymol	89-83-8		321.15–324.15[c]
Salol	118-55-8		314.15[c]

[a] $[\alpha]_D^{20} = -61°$.
[b] From reference 17.
[c] From supplier.
[d] $[\alpha]_D^{20} = -35.3°$.
[e] From reference 28.

Table 2
Physicochemical properties of carvone and carvone-based eutectic mixtures.

Abbreviation	Component 1	Component 2	Mole fraction ($x_{carvone}$)	Viscosity [a] (mPa·s)	Density [b] (g cm^{-3})	Polarity (E_{NR})	Water content (wt%)	
							As is	After water saturation
–	R-(−)-Carvone	–	–	4.38	0.9590	52.95	0.08 (±0.00) [c]	1.53 (±0.01)
CH (0.60)		Heptanoic acid	0.60	3.30	0.9497	52.36	0.13 (±0.00)	3.84 (±0.01)
CO (0.80)		Octanoic acid	0.80	2.61	0.9528	52.46	0.09 (±0.00)	2.76 (±0.01)
CN (0.74)		Nonanoic acid	0.74	3.71	0.9517	52.46	0.10 (±0.00)	2.73 (±0.01)
CD (0.89)		Decanoic acid	0.89	2.18	0.9600	52.65	0.09 (±0.01)	2.04 (±0.00)
CC (0.73)		Camphor	0.73	2.81	0.9579	53.14	0.15 (±0.01)	1.50 (±0.01)
CM (0.76)		Menthol	0.76	3.22	0.9429	52.65	0.11 (±0.00)	1.96 (±0.01)
CB (0.86)		(−)-Borneol	0.86	10.11	0.9614	52.75	0.11 (±0.00)	2.14 (±0.00)
CT (0.50)		Thymol	0.50	5.79	0.9720	50.60	0.09 (±0.00)	2.29 (±0.18)
CS (0.50)		Salol	0.50	3.40	1.0878	52.65	0.07 (±0.00)	0.64 (±0.01)

[a] Measured at 25 ℃ with shear rate 100 s^{-1}.
[b] Measured at 25 ℃.
[c] Mean ± standard deviation (n = 3).

results were visualized using the ggtree package (Yu, 2020). The clustering results are displayed as a hierarchical dendrogram, in which the classification results are indicated.

2.4. Analytical methodologies for physicochemical characterization of solvents

Differential scanning calorimetry (DSC) analysis was carried out using a DSC Q20 (TA Instrument, New Castle, USA) under a nitrogen atmosphere with a flow rate of 50 mL min^{-1}. Mixtures that could be prepared as homogeneous liquids upon heating were used for the measurements. Samples (1–20 mg) were loaded onto a Tzero aluminum pan, with a precision of ± 0.2 mg, and measured through at least three cycles, with a cooling and heating rate of 5 °C min^{-1}. Detailed information on the mole fraction and melting temperature is provided in Table S2.

A TG/DTA (Exstar 7300, Seiko Instrument Inc., Tokyo Japan) was used for thermogravimetric analysis (TGA). The temperature was changed from room temperature to 300 °C at a heating rate of 10 °C min^{-1}.

Fourier transform-infrared spectroscopy (FT-IR) analysis was performed using a Vertex 70 spectrometer, equipped with an attenuated total reflectance probe (Bruker, Billerica, MA, USA). The data were recorded between 500 cm^{-1} and 4000 cm^{-1} at room temperature.

For ^{1}H nuclear magnetic resonance (NMR) spectroscopy analysis, individual compounds and eutectic mixtures were dissolved in $CDCl_3$ at a concentration of 10 v/v%. The measurements were conducted at room temperature using a 700 MHz NMR spectrometer (Bruker, Karlsruhe, Germany).

The solvent viscosity was measured using an ARES-G2 (TA instrument, New Castle, USA) at 25 °C with a shear rate ranging from 1 to 1000 s^{-1}. A shear rate of 100 s^{-1} was used to determine the viscosity in Table 1. The density was measured using a DMA 4500 M density meter (Anton Paar, Graz, Austria) at 25 °C.

The polarity of the pure solvents and mixtures was investigated using a solvatochromic probe, Nile red. The maximum absorption wavelength (λ_{max}) of Nile red dissolved in the tested solvent was measured using a UV/Vis spectrophotometer (Multiskan GO, Thermo Scientific, Vantaa, Finland) at room temperature as previously described (Jin, Jung, Li, Park, & Lee, 2020). E_{NR} was calculated using the following equation:

$$E_{NR} = 28591/\lambda_{max} \tag{4}$$

The water contents of neat solvents and water-saturated solvents were estimated in triplicate using a Karl-Fischer titrimeter (Mettler Toledo Inc., Zurich, Switzerland). Water-saturated solvents were prepared by mixing equal volumes (6 mL for each) of the testing solvents and water, followed by irradiation with ultrasonic waves for 10 min and subsequent centrifugation.

2.5. Biodegradability assessment

Biodegradation of carvone and eutectic mixtures was evaluated using a previously described cultivation method (Kondo, Nishihara, Shimamoto, Watabe, & Fujii, 1988). Briefly, 5 mL of surface water mixture (collected from three places near Sungkyunkwan University, Suwon, Korea) was mixed with 4.9 mL of autoclaved 0.2% peptone solution in a conical tube with a tight cap, and the test species dissolved in acetone was added to the mixture to make a total volume of 10 mL containing 40 mg L^{-1} of test species. A mixture of autoclaved surface water, 0.2% peptone solution, and the test species were used as the negative control. During incubation at 25 ± 2 °C with shaking at 120 rpm, a sample of the solution was taken on days 0, 1, 3, and 5. Absorbance at 610 nm (OD_{610}) was measured using a UV/Vis spectrophotometer to determine bacterial growth, which was not significantly affected by the testing species during the culture period (Fig. S1).

After incubation, the remaining species were extracted using 0.2 mL of hexane, and microextraction was assisted by adding 100 mg of sodium chloride, followed by ultrasonic irradiation for 5 min. After three extractions, the combined extract phases were dried using sodium sulfate. In the case of fatty acid-containing mixtures, the remaining fatty acids were further derivatized to fatty acid methyl esters (FAMEs) by sample acidification to pH 3.0, treatment with 8% methanolic hydrochloride acid at 100 °C for 1 h, and extraction with hexane (Ichihara & Fukubayashi, 2010). The dried extract phases were analyzed by gas chromatography-mass spectrometry (GC–MS), and the separation was performed on an Agilent DB-5MS UI column. The temperature was programmed to hold at 70 °C for 1 min, then increased to 300 °C at 20 °C min^{-1}, and held for 3 min. Except for the injector temperature, which was 250 °C for CS (0.50) and 300 °C for the other species, the other instrumental conditions were set as described (Jeong, Jin, Yoo, Han, Kim, & Lee, 2018). The relative quantity of remaining species was presented as the peak area relative to that of the negative control, with correction of the extracted phase volume.

2.6. Applications of carvone and its mixtures to LLE

Two commercial dyes, hydrophobic rhodamine 6G and hydrophilic Brilliant blue FCF, were added to water at 2.5 mg L^{-1} each, and the same volume of carvone or other eutectic mixtures was added. After vigorous mixing, the mixture was left to phase separate.

Carvone and the mixtures were also used for the extraction of pigments in real samples. The order of addition of solvents to solid samples showed the ability to affect the efficiency of LLE. Specifically, the extraction of the hydrophilic pigments was favored by water followed by organic solvents, while the hydrophobic pigments were better extracted in the opposite order (data not shown). To eliminate this interaction, equal volumes (0.8 mL) of carvone (or eutectic mixture) and water were mixed by vortexing, then 20 mg of laver powder was added, to prevent any preferential extraction of one type over the other. After brief mixing, the mixture was subjected to ultrasound-assisted extraction at ambient conditions for 10 min, then centrifuged at 12,300 × *g* for 10 min. The clear organic and aqueous phases, which contained hydrophobic pigments (carotenoids and chlorophylls) and hydrophilic phycobiliproteins (phycoerythrin and phycocyanin), respectively, were collected using a 500 μL Hamilton syringe (Reno, NV, USA) and diluted with methanol (organic phase) and water (aqueous phase). The diluted extracts were analyzed using a UV/Vis spectrophotometer. The following equations were used to calculate pigment concentrations in accordance with previous studies (Marzorati, Schievano, Ida, & Verotta, 2020):

$$C_{\text{chlorophyll a}}\left(\mu g\ mL^{-1}\right) = 15.65(A_{666}) - 7.34(A_{653}) \tag{5}$$

$$C_{\text{chlorophyll b}}\left(\mu g\ mL^{-1}\right) = 27.05(A_{653}) - 11.21(A_{666}) \tag{6}$$

$$C_{\text{total carotenoids}}\left(\mu g\ mL^{-1}\right) = \left(1000(A_{470}) - 2.86\left(C_{chlorophylla}\right) - 129.2\left(C_{chlorophyllb}\right)\right)/221 \tag{7}$$

$$C_{PC}\ \left(mg\ \ mL^{-1}\right) = 0.154 \times (A_{618} - A_{730}) \tag{8}$$

$$C_{PE}\ \left(mg\ \ mL^{-1}\right) = 0.1247 \times [(A_{564} - A_{730}) - 0.4583 \times (A_{618} - A_{730})] \tag{9}$$

where $A_{wavelength}$ is the absorbance at the designated wavelength (nm), and $C_{pigment}$ is the concentration of a given pigment.

3. Results and discussion

3.1. Preparation of carvone-based eutectic mixtures

The main aim of this study was to use carvone as a bio-based solvent to prepare hydrophobic eutectic mixtures with diverse compounds. For this, several classes of naturally occurring compounds that are reportedly components of hydrophobic eutectic solvents were considered. Straight-chain fatty acid-based eutectic solvents show increasing

hydrophobicity and melting temperature as the carbon chain elongates (Florindo, Romero, Rintoul, Branco, & Marrucho, 2018). In this study, medium-chain aliphatic fatty acids (C_7 to C_{10}) were selected as the model components because they are easy to handle in terms of melting temperature and have frequently been used in the formation of hydrophobic eutectic solvents (Lee, Jung, & Park, 2019). The fatty acids selected were heptanoic acid (enanthic acid), octanoic acid (caprylic acid), nonanoic acid (pelargonic acid), and decanoic acid (capric acid). For the same reasons, camphor, menthol, borneol, and thymol, from which hydrophobic eutectic mixture formation has been reported (Martins et al., 2018), were selected, among a variety of naturally occurring monoterpenes. Although a synthetic chemical, salol (phenyl salicylate), which has many applications including polymer synthesis and pharmaceutical preparation, was also included because it often forms eutectic mixtures with several monoterpenes (Makoś, Słupek, & Gębicki, 2020). The selected compounds are listed in Table 1.

Table S1 shows numerous series of carvone mixtures prepared at various molar ratios via heating and stirring. Many of them remained stable, and are transparent liquids without any crystallization or precipitation while stored at room temperature. All the prepared mixtures were analyzed by DSC to characterize their thermal events, including melting, crystallization, transitions, and glass transitions. The results are indicated in the following SLE phase diagrams.

3.2. SLE phase diagrams of carvone-based eutectic mixtures

The SLE phase diagrams of the carvone mixtures are illustrated in Fig. 1, in which the experimental points of the previously mentioned thermal events are shown by different symbols as a function of mole fractions of carvone ($x_{carvone}$). The curves for ideal eutectic mixtures are not indicated in Fig. 1 for simplicity. Carvone mixtures with heptanoic acid (CH) and nonanoic acid (CN) at several mole fractions displayed phase transitions (triangles) of fatty acids, which is in agreement with previous reports (Schaake, Van Miltenburg, & De Kruif, 1982). All of the fatty acid-based mixtures (Fig. 1a-1d) clearly exhibited a single eutectic temperature (circles), all of which became close to the melting temperatures of the mixtures at certain mole fractions. The ideal curve of the decanoic acid (CD) mixtures (Fig. S2) suggests that carvone could form simple eutectic mixtures with the fatty acid. It is noteworthy that most of the fatty acid-based eutectic mixtures could remain liquid at all the tested mole fractions (0.00–1.00) because of the low melting temperatures of the pure compounds. CD had a slightly narrowed, but still wide liquid window ($x_{carvone}$ of 0.10–1.00). The very wide liquid regions of these fatty acid-based mixtures could make them desirable solvents, which could be tuned to possibly have different properties by simply varying the ratios between carvone and fatty acid.

Fig. 1e–h display the SLE phase diagrams of the terpene-based mixtures. Camphor and borneol showed slightly different thermal behaviors from literature (Martins, Silva, Schaeffer, Abranches, Maximo, Pinho, et al., 2019) in that their melting temperature could not be accurately measured (likely because the melting temperatures are close to their sublimation temperatures) and that their solid phase transitions were not detected (Fig. 1e). Overall, the thermal behaviors of the terpene mixtures varied greatly. The carvone mixtures with menthol (CM) and borneol (CB) presented single eutectic temperatures. It should be noted that the CM mixtures have a liquid window at $x_{carvone}$ of 0.10–1.00 (Fig. 1f), which is much wider compared to many menthol-based eutectic mixtures reported (Martins, et al., 2019). The camphor (CC) mixtures at $x_{carvone}$ of 0.60–0.75 showed neither crystallization nor melting, and they did not apparently freeze at −80 °C. This is similar to

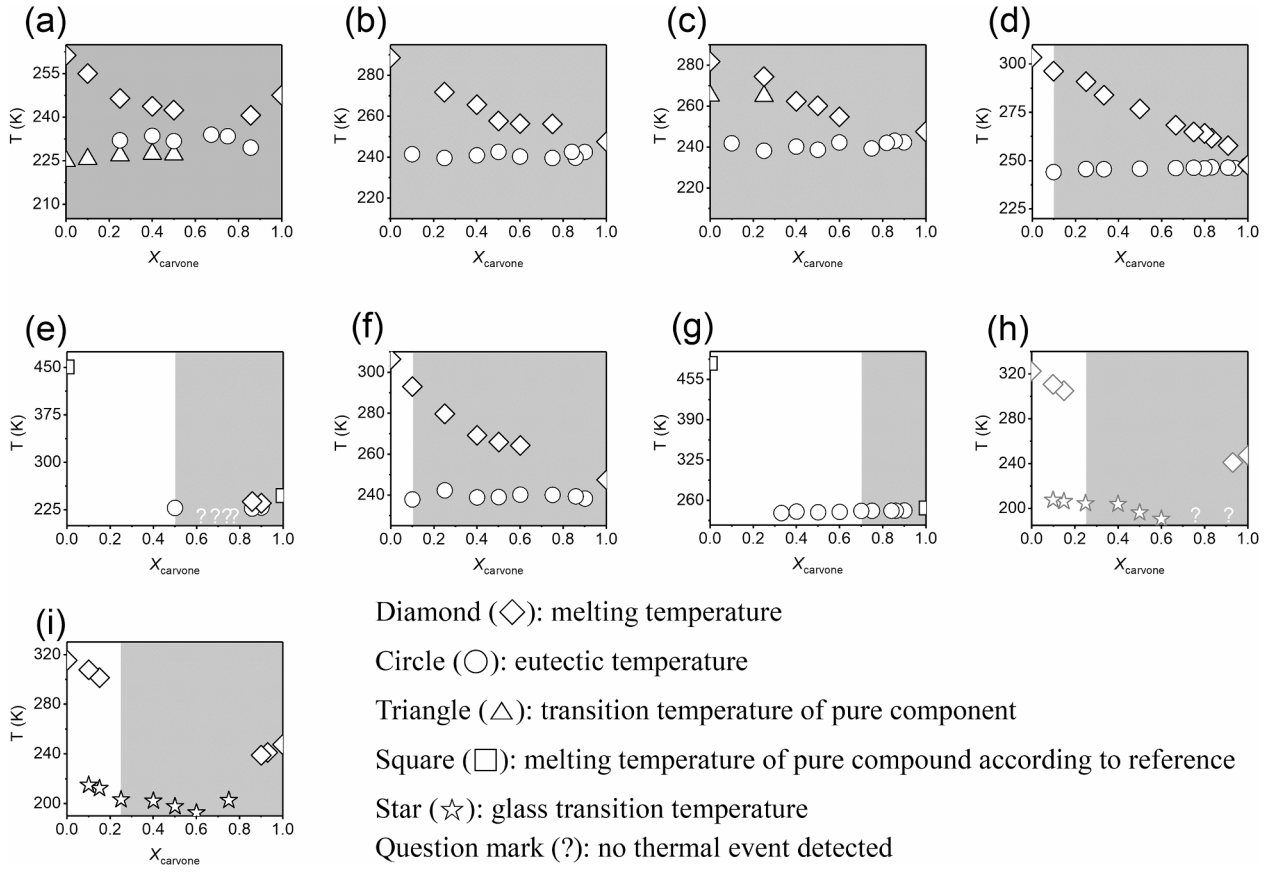


Fig. 1. Solid-liquid equilibrium phase diagrams of the carvone mixtures with heptanoic acid (CH; a), octanoic acid (CO; b), nonanoic acid (CN; c), decanoic acid (CD; d), camphor (CC; e), menthol (CM; f), borneol (CB; g), thymol (CT; h), and salol (CS; i). The gray background indicates the region where the mixtures remain liquid at room temperature.

the mixture of choline chloride and malonic acid that was designated as a DES (Morrison, Sun, & Neervannan, 2009). Despite the lack of exact melting or transition temperatures, the CC mixtures have an apparent liquid range at $x_{carvone} \geq 0.50$. In the case of the thymol (CT) mixtures, no eutectic temperatures were detected, although they presented melting temperatures at $x_{carvone} \leq 0.15$ and $x_{carvone} \geq 0.93$. The CT mixtures also showed glass transition temperatures (stars in Fig. 1h) at $x_{carvone} \leq 0.60$. The apparent melting temperatures of the CT mixtures in $0.25 \leq x_{carvone} \leq 0.90$ deviate greatly from the ideality (Fig. S2). Thus, the CT mixtures may be designated as DESs (Martins, Pinho, & Coutinho, 2019). It may be more appropriate to name the CT mixtures low-transition-temperature mixtures (LTTMs) according to Francisco and coauthors, because they have glass transitions instead of melting temperatures (Francisco, vandenBruinhorst, & Kroon, 2013). Lastly, Fig. 1i shows the salol mixtures (CS). Although salol belongs to a different category from terpenes, CS mixtures behaved similarly to the CT mixtures, including the similar ranges for glass transition and liquid window.

3.3. FT-IR and ^{1}H NMR analyses of carvone-based mixtures

The ideal eutectic point for each mixture series was deduced from the ideal curves (Martins, Pinho, & Coutinho, 2019) and used to prepare the representative eutectic mixtures. They were CH (0.60), CO (0.80), CN (0.74), CD (0.89), CC (0.73), and CB (0.86). The numbers in parentheses indicate $x_{carvone}$ at the ideal eutectic point. The CT and CS mixtures, presenting no eutectic temperatures at any mole fraction, were prepared arbitrarily at $x_{carvone} = 0.5$, yielding CT (0.50) and CS (0.50), respectively. These nine eutectic mixtures were characterized for their physicochemical properties in subsequent studies. The mixture constituents contained functional groups (carbonyl groups, carboxylic acids, and hydroxyl groups). These groups can form hydrogen bonding interactions, which can be recognized by FT-IR and ^{1}H NMR (Hansen, et al., 2021).

Fig. S3 shows the FT-IR spectra of the individual components and their eutectic mixtures. The carbonyl groups in carvone, fatty acids, camphor, and salol were clearly detected in the range of 1600–1700 cm^{-1}. On the other hand, the hydroxyl groups in menthol, borneol, thymol, and salol were barely observed between 3200 and 3550 cm^{-1}, and the O–H vibration in the carboxylic acids was too broad to be identified in the fatty acids. The eutectic mixtures displayed no discernible changes in the wavenumbers or intensities. Thus, the intermolecular interactions between the constituents could not be inferred. The pure compounds and their eutectic mixtures were subjected to ^{1}H NMR analysis, and the spectra are shown in Fig. S4. The ^{1}H NMR spectra showed no new peaks in the eutectic mixtures, confirming that no chemical reaction occurred upon mixture formation (van Osch, Dietz, van Spronsen, Kroon, Gallucci, van Sint Annaland, et al., 2019). No hydroxyl proton peak was observed in pure menthol and borneol, presumably because of its fast exchange with the residual water of the dissolution solvent ($CDCl_3$). Similarly, pure fatty acids did not present carboxylic acid protons. The ^{1}H NMR spectra did not allow exploration of the hydrogen bonding interaction in most mixtures because the functional protons capable of hydrogen-bonding interactions were also invisible in the mixtures (van Osch, et al., 2019). Only thymol displayed the phenolic proton at 4.8 ppm, which was shifted to 5.8 ppm in CT (0.50). This downshift indicates the hydrogen bonding interaction between thymol (phenolic hydroxyl group) and carvone (carbonyl group) (Kang, Namgoong, Son, Kim, Jeong, Jin, et al., 2020). Further NMR measurements by ^{13}C NMR and 2D NMR may help to elucidate these interactions.

3.4. Thermal stability, viscosity, density, polarity, and water contents of carvone and its mixtures

In TGA, a function of temperature versus weight loss is used to determine the evaporation and sublimation points of solvents. The TGA curves of the nine carvone mixtures revealed the onset (T_{onset}) and peak (T_{peak}) temperatures, which are listed in Table S2. Carvone and most mixtures presented a single decay of mass loss, similar to previous reports (Martins et al., 2019). The only exception was CS (0.50), which showed a second plateau. This was similarly observed in several eutectic mixtures including menthol:lidocaine (2:1) (van Osch et al., 2019) and camphor:sobrerol (2.3:1) (Martins et al., 2019) which had interactions between two pure compounds leading to different evaporation temperatures. Many eutectic mixtures showed T_{onset} and T_{peak} values between those of carvone and the counterpart compounds; CM (0.76), CB (0.86), and CT (0.50) presented higher T_{onset} or T_{peak} values than those of the parent compounds. The significantly higher T_{onset} value of CT (0.50) than those of carvone and thymol is likely due to azeotrope formation (Martins et al., 2019), although vapor–liquid equilibrium data are needed for confirmation.

Low viscosity is a desirable solvent property that facilitates easy handling and fast mass transfer during the extraction process. Many of the so-called green solvents, including ILs and DESs, have been criticized for their high viscosity. Hydrophobic eutectic solvents based on menthol and thymol have been reported to have significantly reduced viscosities as low as 11.2 mPa·s at room temperature, while tetraalkyl quaternary ammonium-based solvents were still highly viscous (228–1139 mPa·s) (Lee, Jung, & Park, 2019). As listed in Table 2, the viscosity of carvone was 4.38 mPa·s at 25 °C, and the viscosities of the eutectic mixtures ranged from 2.18 to 10.11 mPa·s. In fact, all the mixtures except for CT (0.50) and CB (0.86) had decreased viscosity compared to carvone. The very low viscosity of carvone and its mixtures could provide these bio-based solvents with great advantages, in order to serve as efficient, readily applicable solvents in industrial extraction processes.

The density of carvone was measured at 0.9590 g cm^{-3}, which is in good agreement with a previous study (Florido, Andrade, Capellini, Carvalho, Aracava, Koshima, et al., 2014). All tested mixtures, except for CS (0.50), were lighter than water (Table 2). Their densities ranged from 0.9429 to 0.9754 g cm^{-3} at room temperature. CS (0.50) was heavier than water, with a density of 1.0878 g cm^{-3}.

Solvent polarity can be characterized in terms of solvatochromic parameters. The multi-parameter scale of Kamlet–Taft (KT) and the $E_T(30)$ of Reichardt's dye are commonly used parameters. However, they could not be measured for carvone and the mixtures, because the innate UV absorbance of carvone interfered with that of KT probes, and because several fatty acids hampered accurate absorbance of Reichardt's dye due to their acidity. Instead, Nile red was used to measure E_{NR} as previously performed (Jin, Jung, Li, Park, & Lee, 2020), and the results are listed in Table 2. In general, the polarity of both the fatty acid- and terpene-based mixtures slightly increased, compared to that of pure carvone ($E_{NR} = 52.95$). The only exception was observed in CC (0.73), which exhibited the highest E_{NR} (53.14) among the tested solvents. In contrast, CT (0.50) was found to be the most polar ($E_{NR} = 50.60$). These results together suggest that the carvone-based eutectic mixtures could be tuned to possess different viscosities, densities, and polarities by varying the type of counterpart.

Limited water solubility can support the hydrophobic character of a solvent, and likely its insignificant loss into water, when the solvent is used with water in LLE. Many terpene-based hydrophobic eutectic mixtures have reasonably low water content after saturation with water. For example, the water-saturated mixtures of thymol with camphor and with borneol contained water at 1.8 wt% and 2.4 wt%, respectively (Martins et al., 2019). On the other hand, fatty acid-based mixtures displayed a varying range of water contents depending on the counterparts; decanoic acid mixtures with tetraalkyl ammonium salts could range in 1.8–6.9 wt% of water (van Osch, Zubeir, van den Bruinhorst, Rocha, & Kroon, 2015), while those with lidocaine could be as high as 20.6 wt% (van Osch, et al., 2019). The water content in neat carvone was measured 0.08 wt%, and the eutectic mixtures before water saturation contained 0.07–0.48 wt% of water (Table 2). After saturation

with water, all the solvents contained relatively low levels of water (0.64–3.84 wt%). The volumes of the two phases did not change significantly after saturation with water. When characterizing CT (0.50), the most polar solvent among the tested eutectic mixtures, the NMR spectra were further examined before and after water saturation. The molar ratios of carvone to thymol did not appear to change significantly after the water saturation (Fig. S4), while the water contents increased from 0.09 wt% to 2.29 wt%. It is inferred that the mutual solubility between water and CT (0.50) is negligible, with a low chance of losing CT (0.50) into the aqueous phase, and that the other mixtures are likely to behave similarly to CT (0.50). In this sense, carvone-based mixtures can make sustainable, hydrophobic solvents for LLE.

3.5. *Thermodynamic investigation of carvone mixtures using COSMO-RS*

Thermodynamic examinations were performed to further understand the molecular interactions in carvone mixtures. COSMO-RS allowed the transfer of the 3D distribution of polarization charges on each molecule to the surface composition function (σ-profile) and chemical potential (σ-potential). The resulting surface polarity of carvone and the mixture is shown in Fig. 2. The screening charge density distribution can be divided into HBD (negative charge density), non-polar (neutral charge density), and HBA (positive charge density) regions by two vertical lines of σ at −0.0084 e/Å^2 and +0.0084 e/Å^2 in the σ-profile. Carvone displayed a broad, prominent peak in the non-polar region, supporting its hydrophobic character. It also showed a weakly polar character, as represented by a peak at +0.013 e/Å^2, which is ascribed to the carbonyl group. Thus, carvone can presumably promote hydrogen-bonding interactions as HBA, although it lacks peaks in the HBD region. Compared to neat carvone, the mixtures showed similar intensity in the non-polar region and retained the promoting capability of hydrogen-bonding interactions in the HBA region. Notably, CT (0.50)

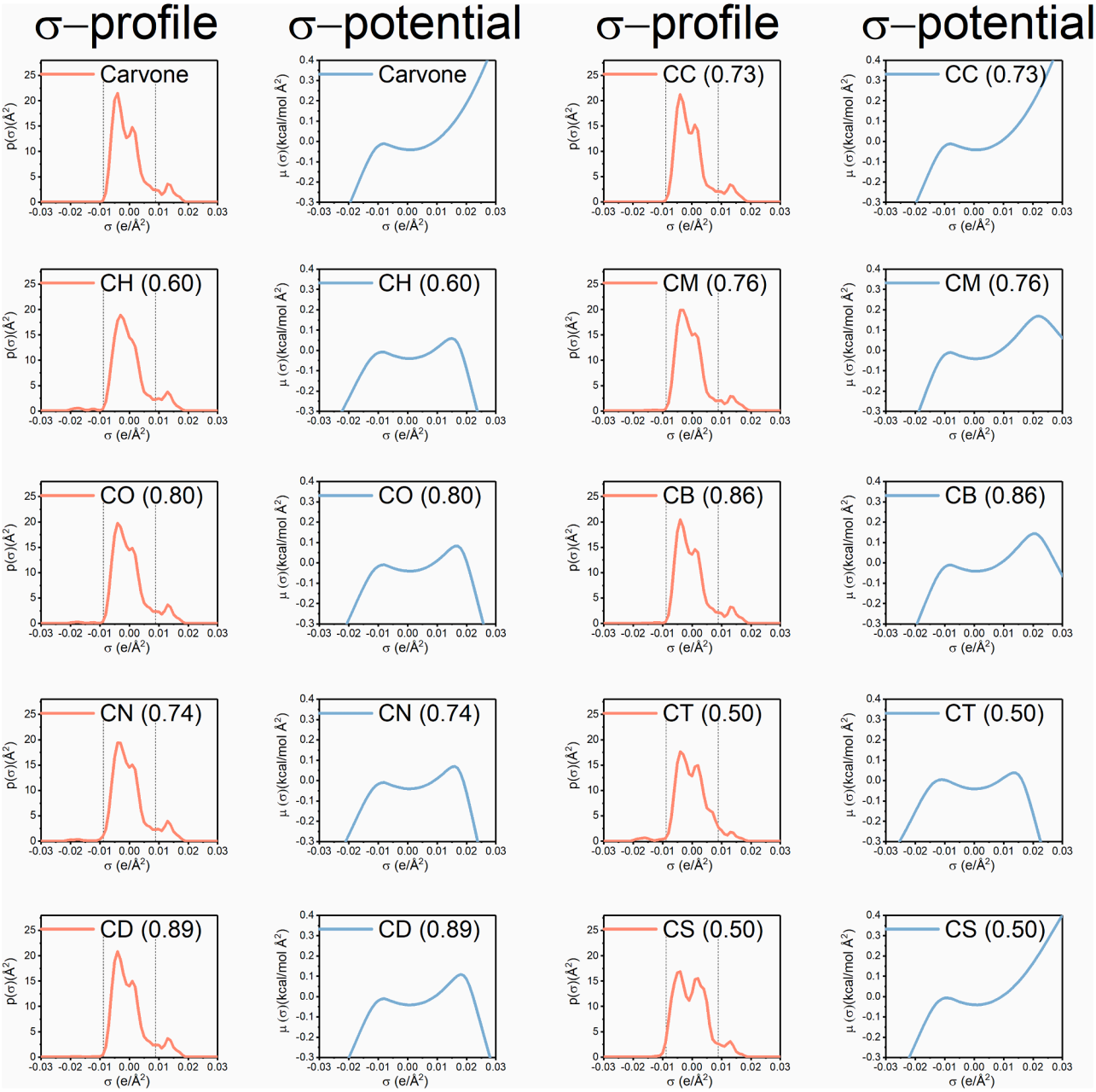


Fig. 2. COSMO-RS-derived σ-profiles (orange) and σ-potentials (blue) of carvone and its mixtures, CH (0.60), CO (0.80), CN (0.74), CD (0.89), CC (0.73), CM (0.76), CB (0.86), CT (0.50), and CS (0.50). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

exhibited the hydrogen-donating capacity provided by thymol, as indicated by discernible peaks in the HBD region (charge density of −0.016 e/Å^2). This agrees well with the NMR result, which indicated the intermolecular hydrogen bonding between the carbonyl group of carvone and the phenolic hydroxyl proton of thymol (Fig. S4).

A more negative σ-potential indicates a higher affinity to HBAs or HBDs. The fatty acid-based mixtures generally had lower σ-potentials than the two saturated terpene-based mixtures, CM (0.76) and CB (0.86). In addition, the terpene-based mixtures exhibited different patterns. For instance, CT (0.50) presented the lowest value in the HBA region, which was attributed to the hydroxy group of thymol, whereas CC (0.73) presented the largest values in the HBA region. These results are mostly in agreement with the polarity results in Table 1, in which CT (0.50) and CC (0.73) were the most and least polar solvents, respectively.

3.6. Classification of carvone and its mixtures as substitutes for conventional solvents

The selection of solvents is a critical task in the extraction process, and is under current pressure towards green chemistry. Many physicochemical properties, in combination with multivariate statistical analyses, have been employed to appropriately classify numerous solvents. Although KT parameters are frequently used as a quantitative measure of solvent properties, they are not available for all solvents in the literature, and they cannot be measured for certain solvents, as was the case in this study. Hansen solubility parameters are often employed for the rational selection of solvents; however, the calculated values may differ depending on the methods (Sanchez-Camargo, Bueno, Parada-Alfonso, Cifuentes, & Ibanez, 2019). In this context, recent studies have adopted machine learning and deep learning algorithms to predict the physicochemical properties of solvents (Hansen, et al., 2021). Aubry et al. used a COSMO-RS model to calculate σ-potentials of 153 classical solvents and applied principal component analysis (PCA) for dimension reduction of datasets, along with k-means for clustering of the solvents (Durand, Molinier, Kunz, & Aubry, 2011). Their approach showed that non-empirical information derived solely from chemical structures could potentially be used to cluster solvents and help select replacement solvents (Moity, Durand, Benazzouz, Pierlot, Molinier, & Aubry, 2012).

As such, we attempted to place the new solvents, carvone and its mixtures, among various classes of conventional solvents. For this, σ-potentials of 150 solvents (Table S3) were calculated in reference to

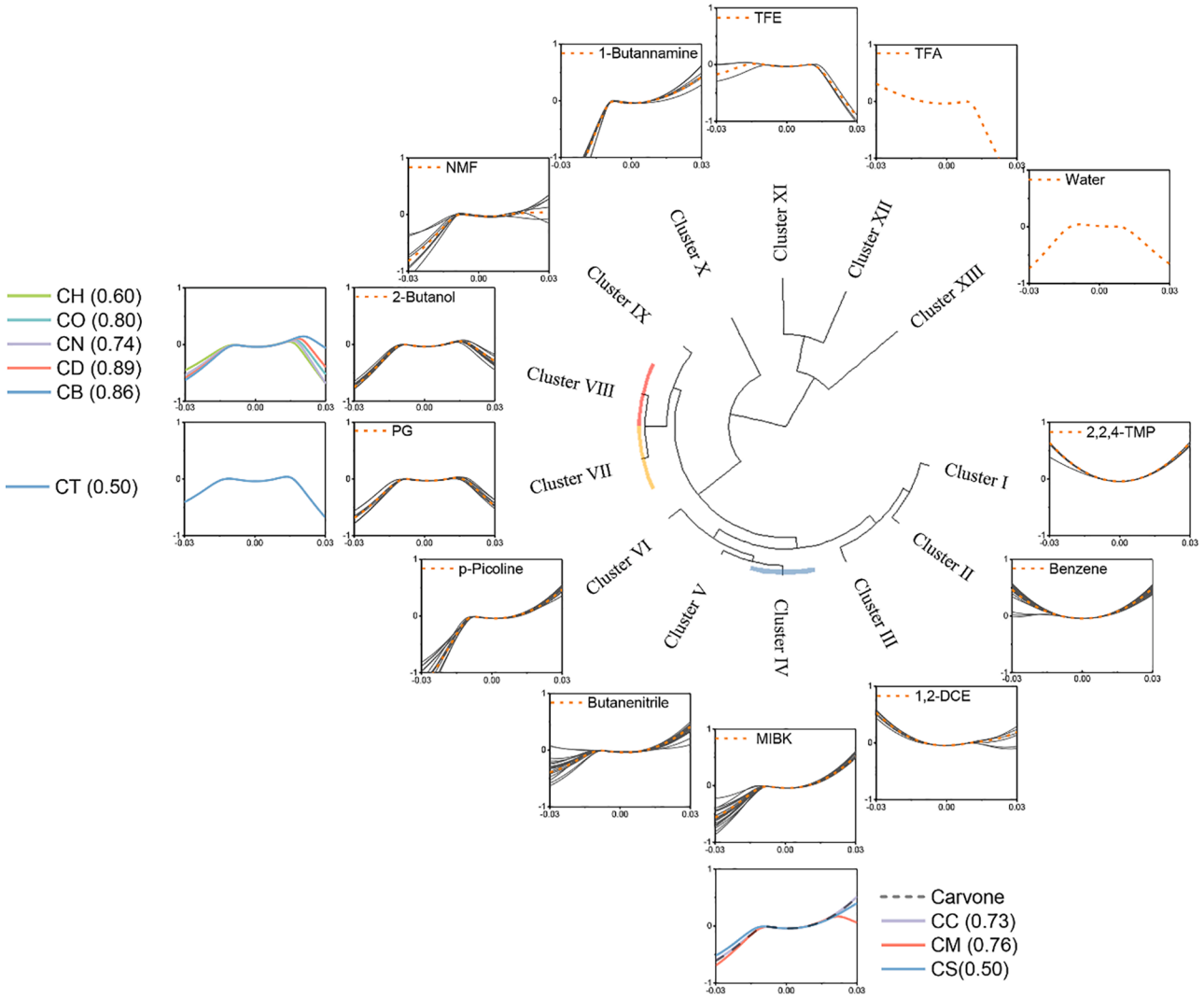


Fig. 3. Clustering of 150 conventional solvents and classification of carvone and its mixtures among 13 clusters, based on the σ-potentials. For each cluster, σ-potentials of conventional solvents in gray solid line are overlaid with an exemplar in orange dashed line. The σ-potentials of carvone and its mixtures are displayed in color in the clusters where they belong to. For descriptions of the carvone mixtures, please see Table 1. Solvent abbreviations: 1,2-DCE, *cis*-1,2-dichloroethene; 2,2,4-TMP, 2,2,4-trimethylpentane; MIBK, methyl isobutyl ketone; NMF, *N*-methylformamide; PG, 1,2-propylene glycol; TFA, trifluoroacetic acid; TFE, 2,2,2-trifluoroethanol. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

the study by Aubry et al. (Durand, Molinier, Kunz, & Aubry, 2011). All told, 61 variables for each of the 150 solvents were acquired; after data scaling, the datasets were directly applied to a clustering algorithm without dimension reduction. The AP algorithm was selected for clustering, which is based on message passing between data points (Frey & Dueck, 2007). This choice had its advantages, in that it requires no cluster numbers and provides representatives or exemplars of clusters. As shown in Fig. 3, the solvents were divided into 13 clusters (clusters I–XIII). The hierarchy of the clusters is displayed in a dendrogram, and the σ-potential of an exemplar solvent is shown by the orange dashed line among the other solvents in each cluster. The exemplars for clusters I to XIII are 2,2,4-trimethylpentane, benzene, *cis*-1,2 dichloroethene, methyl isobutyl ketone, butanenitrile, *p*-picoline, 1,2-propylene glycol, 2-butanol, *N*-methylformamide, 1-butanamine, 2,2,2-trifluroethanol, trifluoroacetic acid, and water. The assigned clusters are indicated for all the solvents in Table S3. The present study produced a slightly larger number of clusters than in the literature (10 clusters) (Durand, Molinier, Kunz, & Aubry, 2011). The difference is largely attributed to the different clustering results for the polar protic solvents (clusters X – XIII), which were distinctly clustered from the other solvents.

In the second step, the 150 datasets with cluster labels as a training set were applied to SVMs, which have been widely used for classification and regression analysis and have shown to work effectively in high-dimensional dataset analysis. The obtained classification model was used to predict the clusters of carvone and its mixtures. As displayed in Fig. 3, carvone was classified as cluster IV; however, its mixtures could be placed in different clusters depending on the partnering components. CC (0.73), CM (0.76), and CS (0.5) were classified together with carvone in cluster IV, which is represented by methyl isobutyl ketone and contains ketone- and ether-based solvents. It is reasonable to regard cluster IV as analogous to the aprotic dipolar cluster (Durand, Molinier, Kunz, & Aubry, 2011; Moity, Durand, Benazzouz, Pierlot, Molinier, & Aubry, 2012). It is also reasonable to presume that carvone and the three carvone mixtures may substitute for the conventional organic solvents in cluster IV. All of the fatty acid mixtures were classified into cluster VIII, which largely consists of aliphatic mono-alcohols, as represented by 2-butanol. CB (0.86) was unexpectedly observed in cluster VIII. CT (0.50), which showed the highest solvatochromic polarity, was found in cluster VII. Cluster VII, represented by 1,2-propylene glycol, includes various polar protic solvents with mono- to polyalcohol groups. The classification results above show that our classification model built from the σ-potentials enables reasonable prediction of the solvent clusters of carvone and its mixtures. Moreover, the results imply that carvone may be tuned by preparing eutectic mixtures with diverse compounds to serve as green replacement solvents.

3.7. Biodegradability of carvone and its mixtures

Biodegradability is an important aspect of solvents, in order to ensure greenness (Clarke, Tu, Levers, Bröhl, & Hallett, 2018). Despite a limited number of detailed assessments, the biodegradability of DESs has been assumed simply based on properties of nature-derived constituents. Moreover, most studies have focused on hydrophilic solvents. Carvone is known to be readily biodegradable in a closed bottle test (OECD301D), and most of its counterparts are reported to be biodegradable. Nonetheless, it is still important to examine whether the eutectic mixtures will be as biodegraded as their pure compounds are. Here, the biodegradability of all the mixtures was carefully examined using a previously described cultivation method (Kondo, Nishihara, Shimamoto, Watabe, & Fujii, 1988). The mixtures were monitored for 0–5 days by extraction of the remaining compounds and subsequent GC–MS analysis. The experimental conditions for the liquid-phase microextraction were optimized using a one-variable-at-a-time strategy, including the type of extraction solvents (hexane, chloroform, and toluene), ultrasonic irradiation time (0–5 min), and amount of sodium chloride (0–200 mg) in reference to literature (Chiu & Kuo, 2020; Lee, Jung, & Park, 2019) (data not shown). Each test species incubated with the autoclaved inoculum (no microbe) was used as the negative control, and neat carvone was used as the positive control.

The remaining amounts of test species relative to day 0 are listed in Table S4. After five days of incubation, half of the carvone remained intact, while the degradation was found to involve reduction of carvone to dihydrocarvone (data not shown). In the case of the eutectic mixtures, most were degraded in similar pathways, but sometimes at different degradation speeds. Specifically, faster degradation was observed in CH (0.60) and CS (0.50), and slower degradation was observed in CB (0.86) and CT (0.50) (Table S4). Collectively, it is concluded that carvone and carvone mixtures are reasonably biodegradable, indicating that they are suitable as novel green solvents.

3.8. Applications of carvone and its mixtures to LLE

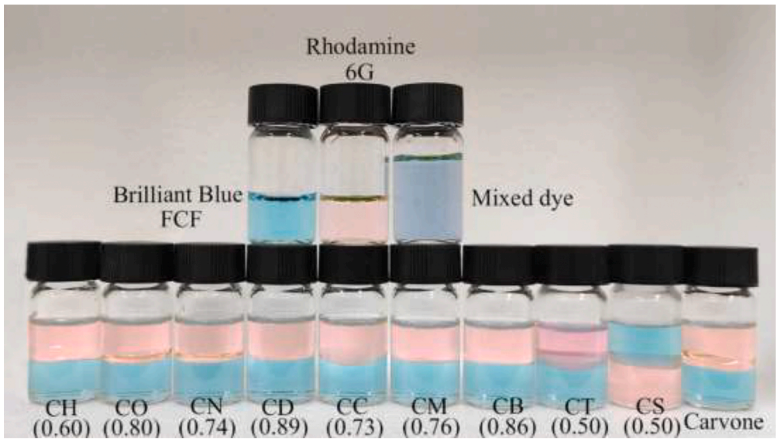
Carvone and the eutectic mixtures were applied to aqueous solutions containing two model dyes, rhodamine 6G (red) and brilliant blue FCF (blue). Fig. 4a shows that all these solvents were capable of forming a distinct phase from water. The red hydrophobic dye could be preferentially partitioned into the organic phase, which resided on the top of the water, except for in CS (0.50), which is uniquely denser than water. Next, carvone and the mixtures were applied to the LLE of valuable compounds from a real sample, laver. Laver is a red macroalgae that is popularly consumed as a food source and contains a wide variety of compounds with different activities and properties. In this study, major functional pigments in laver, two hydrophilic phycobiliproteins PC and PE, and three hydrophobic pigments, carotenoids, chlorophyll *a*, and chlorophyll *b*, were targeted for extraction due to their distinct colors and high economic value.

Fig. 4b and 4c show that the hydrophobic and hydrophilic pigments could be extracted into the organic and aqueous phases, respectively, with varying efficiencies. Carvone and the mixtures that were placed in cluster IV, along with CB (0.86) in cluster VIII, generally displayed similar extraction yields in both the organic and aqueous phases. CS (0.50) produced the highest yields of PC and PE and displayed a significantly lower water content, despite a very similar polarity to CM (0.76) and CB (0.86) (Table 2). CT (0.50), which had a significantly lower E_{NR} value compared to the other terpene mixtures and was uniquely located with polar protic solvents in cluster VII, showed the lowest yields for the hydrophobic pigments. These observations imply that the solvent polarity and the mutual solubility between the solvents affect the extraction efficiencies for the two classes of pigments studied.

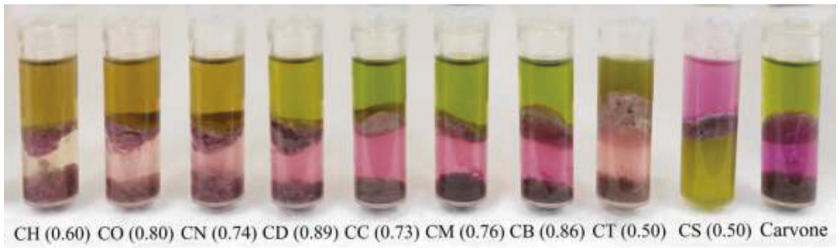
The fatty acid mixtures, which were all placed in cluster VIII, showed different extraction results. These mixtures showed increasing extraction yields, for both the hydrophobic and hydrophilic pigments, as the carbon chain of the fatty acid elongated, with the exception of CH (0.60). Results show that CH (0.60) extracted the largest amounts of carotenoids and chlorophyll *a*, while also leaving no PC or PE in the aqueous phase. These results are presumably associated with the solvent polarity of the organic phase and the acidity of the aqueous phase. Specifically, the extraction yields for the hydrophobic pigments increased as the solvent polarity decreased (Table 2), and the yields for the hydrophilic pigments increased with decreasing acidity (pH 3.14 in CH (0.60) to pH 3.43 in CN (0.89)). The solubility and stability of PC and PE, whose pI values are around 4.1 to 4.2, can be deteriorated by acid (Orta-Ramirez, Merrill, & Smith, 2000). Indeed, the purple aggregates that are indicative of PE and PC precipitates were the most evident in the case of CH (0.60) and less evident in the others (Fig. 4b). It is inferred that the fatty acids in the eutectic mixtures, especially heptanoic acid in CH (0.60), which exhibited the highest water content (3.8 wt% when saturated with water) was slightly dissolved in water to acidify the solution.

The above application experiments suggest that carvone itself could serve as a hydrophobic bio-based solvent in LLE. More importantly, the extraction performance could be modified to have different selectivity

(a)



(b)



(c)

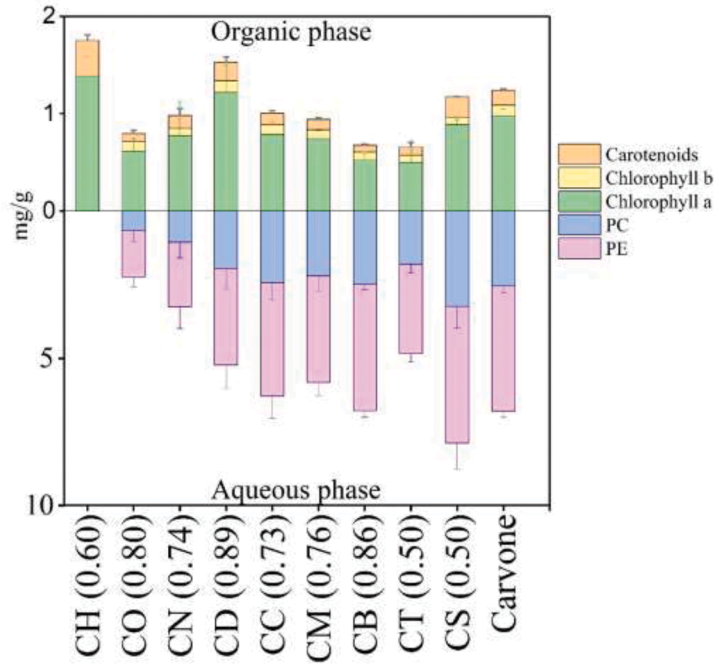


Fig. 4. Applications of carvone and its mixtures as the extraction solvents in liquid–liquid extraction. (a) Partition of hydrophobic (rhodamine 6G; red) and hydrophilic (Brilliant blue FCF; blue) dyes between the organic and aqueous phases. (b) Liquid-liquid extraction of laver containing hydrophobic pigments (greenish phase containing carotenoids and chlorophylls) and hydrophilic pigments (purplish phase containing PC and PE). (c) Extraction yields of individual pigments from the laver extraction. For descriptions of the carvone mixtures, please see Table 1. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

and efficiency by forming eutectic mixtures of carvone with various compounds.

4. Conclusions

This study investigated carvone as a potential bio-based hydrophobic solvent. Carvone can form eutectic mixtures with fatty acids, terpenes, and a benzoate ester. The carvone mixtures showed a wide liquid window, with a low viscosity (2.61–10.11 mPa·s). Carvone and its mixtures were immiscible with water (0.6–3.8 wt% water after water saturation) and reasonably biodegradable. Their physicochemical properties, including density, polarity, and thermal stability, differed depending on

the composition of the mixture. These results imply that the solvent properties may be tuned by changing the counterparts to carvone and the molar ratios between the components. Although hydrogen bonding interactions in the carvone mixtures could not be evidenced by FT-IR and ^{1}H NMR, they were thermodynamically supported in the σ-profiles calculated by COSMO-RS. Application of the σ-potentials to AP and SVMs modeling suggests that neat carvone and the camphor and menthol mixtures may substitute for aprotic dipolar solvents with ketone or ether groups (cluster IV). In the same context, the fatty acid mixtures and the thymol mixture were suggested to be substitution solvents for aliphatic mono-alcohols (cluster VIII) and polar protic mono- to poly-alcohols (cluster VII), respectively. All of the solvents were effective in partitioning the hydrophobic model dye in LLE and presented varying efficiency and selectivity in the extraction of natural hydrophobic pigments from laver. Taken together, carvone is proposed here as a new bio-based source of hydrophobic designer solvents, through the formation of eutectic mixtures with various counterparts. The produced carvone mixtures can serve as low-viscosity, tunable solvents for liquid–liquid extractions involving water.

CRediT authorship contribution statement

Ke Li: Conceptualization, Software, Formal analysis, Investigation, Writing – original draft. **Jin Hyun Nam:** Software, Formal analysis, Validation. **Seulgi Kang:** Resources, Visualization. **Yuli Liu:** Resources. **Jeongmi Lee:** Conceptualization, Writing – review & editing, Supervision, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This study received financial support from a research grant from the National Research Foundation of Korea (No. 2020R1A2C1014006). The authors would like to acknowledge Mr. Zhenchuan Tian for technical advice on the DSC measurements.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.foodchem.2021.131630.

References

Abbott, A. P., Harris, R. C., Ryder, K. S., D'Agostino, C., Gladden, L. F., & Mantle, M. D. (2011). Glycerol eutectics as sustainable solvent systems. *Green Chemistry, 13*(1), 82–90.

Boutekedjiret, C., Vian, M. A., & Chemat, F. (2014). Terpenes as green solvents for natural products extraction. In *Alternative Solvents for Natural Products Extraction* (pp. 205–219). Springer.

Chiu, H.-H., & Kuo, C.-H. (2020). Gas chromatography-mass spectrometry-based analytical strategies for fatty acid analysis in biological samples. *Journal of Food and Drug Analysis, 28*(1), 60–73.

Clarke, C. J., Tu, W.-C., Levers, O., Bröhl, A., & Hallett, J. P. (2018). Green and Sustainable Solvents in Chemical Processes. *Chemical Reviews, 118*(2), 747–800.

Decarvalho, C., & Dafonseca, M. (2006). Carvone: Why and how should one bother to produce this terpene. *Food Chemistry, 95*(3), 413–422.

Durand, M., Molinier, V., Kunz, W., & Aubry, J.-M. (2011). Classification of Organic Solvents Revisited by Using the COSMO-RS Approach. *Chemistry-a European Journal, 17*(18), 5155–5164.

Florido, P. M., Andrade, I. M. G., Capellini, M. C., Carvalho, F. H., Aracava, K. K., Koshima, C. C., … Gonçalves, C. B. (2014). Viscosities and densities of systems involved in the deterpenation of essential oils by liquid-liquid extraction: New UNIFAC-VISCO parameters. *The Journal of Chemical Thermodynamics, 72*, 152–160.

Florindo, C., Branco, L. C., & Marrucho, I. M. (2019). Quest for Green-Solvent Design: From Hydrophilic to Hydrophobic (Deep). *Eutectic Solvents., 12*(8), 1549–1559.

Florindo, C., Romero, L., Rintoul, I., Branco, L. C., & Marrucho, I. M. (2018). From Phase Change Materials to Green Solvents: Hydrophobic Low Viscous Fatty Acid-Based Deep Eutectic Solvents. *Acs Sustainable Chemistry & Engineering, 6*(3), 3888–3895.

Francisco, M., vandenBruinhorst, A., & Kroon, M. C. (2013). Low-Transition-Temperature Mixtures (LTTMs). *A New Generation of Designer Solvents., 52*(11), 3074–3085.

Frey, B. J., & Dueck, D. (2007). Clustering by passing messages between data points. *Science, 315*(5814), 972–976.

Gallis, H. E., Bougrioua, F., Oonk, H. A. J., vanEkeren, P. J., & vanMiltenburg, J. C. (1996). Mixtures of d- and l-carvone.1. Differential scanning calorimetry and solid-liquid phase diagram. *Thermochimica Acta, 274*, 231–242.

García, J. I., García-Marín, H., & Pires, E. (2014). Glycerol based solvents: Synthesis, properties and applications. *Green Chemistry, 16*(3), 1007–1033.

Gu, Y. L., & Jerome, F. (2013). Bio-based solvents: An emerging generation of fluids for the design of eco-efficient processes in catalysis and organic chemistry. *Chemical Society Reviews, 42*(24), 9550–9570.

Hansen, B. B., Spittle, S., Chen, B., Poe, D., Zhang, Y., Klein, J. M., … Sangoro, J. R. (2021). Deep eutectic solvents: A review of fundamentals and applications. *Chemical Reviews, 121*(3), 1232–1285.

Ichihara, K., & Fukubayashi, Y. (2010). Preparation of fatty acid methyl esters for gas-liquid chromatography. *Journal of Lipid Research, 51*(3), 635–640.

Jeong, K. M., Jin, Y., Yoo, D. E., Han, S. Y., Kim, E. M., & Lee, J. (2018). One-step sample preparation for convenient examination of volatile monoterpenes and phenolic compounds in peppermint leaves using deep eutectic solvents. *Food Chemistry, 251*, 69–76.

Jin, Y., Jung, D., Li, K., Park, K., & Lee, J. (2020). Mixing of menthol-based hydrophobic deep eutectic solvents as a novel method to tune their properties. *Journal of Molecular Liquids, 301*, 112416. https://doi.org/10.1016/j.molliq.2019.112416

Kang, S., Namgoong, H., Son, W. S., Kim, Y. D., Jeong, K. M., Jin, Y., … Lee, J. (2020). Insights into the Vastly Different Effects of Eutectic Solvents on the Stability of Phenolic Compounds. *Journal of Physical Chemistry Letters, 11*(13), 5268–5272.

Klamt, A. (2005). *COSMO-RS: From quantum chemistry to fluid phase thermodynamics and drug design*. Elsevier.

Kondo, M., Nishihara, T., Shimamoto, T., Watabe, K., & Fujii, M. (1988). Screening Test Method for Degradation of Chemicals in Water. A Simple and Rapid Method for Biodegradation Test (Cultivation Method). *Eisei kagaku, 34*(2), 115–122.

Lange, B. M. (2015). Biosynthesis and Biotechnology of High-Value p-Menthane Monoterpenes, Including Menthol, Carvone, and Limonene. In J. Schrader, & J. Bohlmann (Eds.), *Biotechnology of Isoprenoids* (pp. 319–353). Cham: Springer International Publishing.

Lee, J., Jung, D., & Park, K. (2019). Hydrophobic deep eutectic solvents for the extraction of organic and inorganic analytes from aqueous environments. *Trac-Trends in Analytical Chemistry, 118*, 853–868.

Makoś, P., Słupek, E., & Gębicki, J. (2020). Hydrophobic deep eutectic solvents in microextraction techniques–A review. *Microchemical Journal, 152*, 104384. https://doi.org/10.1016/j.microc.2019.104384

Martins, M. A. R., Silva, L. P., Schaeffer, N., Abranches, D. O., Maximo, G. J., Pinho, S. P., & Coutinho, J. A. P. (2019). Greener Terpene-Terpene Eutectic Mixtures as Hydrophobic Solvents. *Acs Sustainable Chemistry & Engineering, 7*(20), 17414–17423.

Martins, M. A. R., Crespo, E. A., Pontes, P. V. A., Silva, L. P., Bülow, M., Maximo, G. J., … Coutinho, J. A. P. (2018). Tunable Hydrophobic Eutectic Solvents Based on Terpenes and Monocarboxylic Acids. *Acs Sustainable Chemistry & Engineering, 6*(7), 8836–8846.

Martins, M. A. R., Pinho, S. P., & Coutinho, J. A. P. (2019). Insights into the Nature of Eutectic and Deep Eutectic Mixtures. *Journal of Solution Chemistry, 48*(7), 962–982.

Marzorati, S., Schievano, A., Idà, A., & Verotta, L. (2020). Carotenoids, chlorophylls and phycocyanin from Spirulina: Supercritical CO2 and water extraction methods for added value products cascade. *Green Chemistry, 22*(1), 187–196.

Moity, L., Durand, M., Benazzouz, A., Pierlot, C., Molinier, V., & Aubry, J.-M. (2012). Panorama of sustainable solvents using the COSMO-RS approach. *Green Chemistry, 14*(4), 1132–1145.

Morrison, H. G., Sun, C. C., & Neervannan, S. (2009). Characterization of thermal behavior of deep eutectic solvents and their potential as drug solubilization vehicles. *International Journal of Pharmaceutics, 378*(1-2), 136–139.

Orta-Ramirez, A., Merrill, J. E., & Smith, D. M. (2000). pH affects the thermal inactivation parameters of R-phycoerythrin from Porphyra yezoensis. *Journal of Food Science, 65*(6), 1046–1050.

Ozturk, B., Esteban, J., & Gonzalez-Miquel, M. (2018). Deterpenation of Citrus Essential Oils Using Glycerol-Based Deep Eutectic Solvents. *Journal of Chemical and Engineering Data, 63*(7), 2384–2393.

Pangestuti, R., & Kim, S. K. (2011). Biological activities and health benefit effects of natural pigments derived from marine algae. *Journal of Functional Foods, 3*(4), 255–266.

Pereira, C. V., Silva, J. M., Rodrigues, L., Reis, R. L., Paiva, A., Duarte, A. R. C., & Matias, A. (2019). Unveil the anticancer potential of limomene based therapeutic deep eutectic solvents. *Scientific Reports, 9*(1), 14926.

Ribeiro, B. D., Florindo, C., Iff, L. C., Coelho, M. A. Z., & Marrucho, I. M. (2015). Menthol-based Eutectic Mixtures: Hydrophobic Low Viscosity Solvents. *Acs Sustainable Chemistry & Engineering, 3*(10), 2469–2477.

Sánchez-Camargo, A. D. P., Bueno, M., Parada-Alfonso, F., Cifuentes, A., & Ibáñez, E. (2019). Hansen solubility parameters for selection of green extraction solvents. *Trac-Trends in Analytical Chemistry, 118*, 227–237.

Schaake, R., Van Miltenburg, J., & De Kruif, C. J. T. J. o. C. T. (1982). Thermodynamic properties of the normal alkanoic acids I. Molar heat capacities of seven odd-numbered normal alkanoic acids. *14*(8), 763-769.

van Osch, D. J. G. P., Dietz, C. H. J. T., van Spronsen, J., Kroon, M. C., Gallucci, F., van Sint Annaland, M., & Tuinier, R. (2019). A Search for Natural Hydrophobic Deep Eutectic Solvents Based on Natural Components. *Acs Sustainable Chemistry & Engineering, 7*(3), 2933–2942.

van Osch, D. J. G. P., Zubeir, L. F., van den Bruinhorst, A., Rocha, M. A. A., & Kroon, M. C. (2015). Hydrophobic deep eutectic solvents as water-immiscible extractants. *Green Chemistry, 17*(9), 4518–4521.

Yu, G. (2020). Using ggtree to Visualize Data on Tree-Like Structures. *Current Protocols in Bioinformatics, 69*(1). https://doi.org/10.1002/cpbi.v69.110.1002/cpbi.96